

Quantitative Aptitude

Number System

STUDY NOTES

LCM and HCF



LCM and HCF

LCM:

The **Least Common Multiple (LCM)** of two or more integers is the smallest positive integer that is a multiple of each of the numbers. In other words, it is the smallest number that can be divided evenly by each of the given numbers.

Common Multiples:

1. For two numbers, common multiples are the numbers that are multiples of both numbers. For example, the common multiples of 4 and 6 are 12, 24, 36, etc.
2. The LCM is the smallest of these common multiples.

Properties of LCM

3. **Non-zero Requirement:** The LCM of any set of numbers is always a non-zero integer.
4. **LCM of a Number and 1:** The LCM of any number and 1 is the number itself (e.g., LCM of 5 and 1 is 5).
5. **Commutative Property:** The LCM of two numbers does not depend on the order of the numbers. Thus, $\text{LCM}(a, b) = \text{LCM}(b, a)$.
6. **Associative Property:** The LCM of three numbers can be found as $\text{LCM}(a, \text{LCM}(b, c))$, and this property allows us to find the LCM of multiple numbers iteratively.

Finding the LCM

There are several methods to find the LCM of two or more numbers:

1. Listing Multiples Method

This is a simple but less efficient method, especially for large numbers. It involves listing the multiples of each number until a common multiple is found.

Example 1: Find the LCM of 4 and 6.

Solution:

Multiples of 4: 4, 8, 12, 16, 20, 24, ...

Multiples of 6: 6, 12, 18, 24, 30, ...

Common multiples of 4 and 6: 12, 24, ...

LCM = 12 (the smallest common multiple).

2. Prime Factorization Method

In this method, each number is expressed as a product of prime factors, and the LCM is obtained by taking the highest power of each prime factor present in the factorization.

Example 2: Find the LCM of 8 and 12.

Solution:

Prime factorization of 8: $8 = 2^3$

Prime factorization of 12: $12 = 2^2 \times 3^1$

For the LCM, take the highest power of each prime:

Power of 2: 2^3

Power of 3: 3^1

LCM = $2^3 \times 3 = 8 \times 3 = 24$

Example 3: What is the least common multiple (LCM) of 2, 3, and 8?

Solution:

Calculation:

Find the prime factors of each number:

$$2 = 2$$

$$3 = 3$$

$$8 = 2 \times 2 \times 2 = 2^3$$

Identify the highest power of each prime factor: 2^3 and 3^1 .

Multiply the highest powers together to find the LCM:

$$\text{LCM} = 2^3 \times 3^1 = 8 \times 3 = 24.$$

Therefore, the LCM of 2, 3, and 8 is 24.

3. Division Method

The division method is efficient for finding the LCM of several numbers. This method involves dividing the numbers by their common prime factors until only 1s remain.

Begin by writing all the specified numbers on a line separated by a comma. Then, divide these numbers by the least common prime factors, such as 2, 3, 5, 7, and 11..., and put the quotients right below the numbers, separated by a comma.

If a number is not divisible by a prime factor, write it below itself. Continue dividing by higher prime factors until the quotient on the last line is 1.

Calculate the product of all prime factors used to divide the numbers in different phases. This product will be the LCM of the provided numbers.

Example 4: Find the LCM of 6, 8, and 12.

Solution:

Write the numbers in a row: 6, 8, 12

Divide by the smallest prime that can divide at least one of the numbers (beginning with 2):

$$2: 6 \div 2 = 3, 8 \div 2 = 4, 12 \div 2 = 6 \Rightarrow \text{New row: } 3, 4, 6$$

$$2: 4 \div 2 = 2, 6 \div 2 = 3 \Rightarrow \text{New row: } 3, 2, 3$$

$$3: 3 \div 3 = 1, 3 \div 3 = 1, \Rightarrow \text{New row: } 1, 2, 1$$

2: $2 \div 2 = 1 \Rightarrow$ New row: 1, 1, 1

Multiply all the divisors: $2 \times 2 \times 3 \times 2 = 24$

LCM = 24

Example 5: Find the LCM of 24, 36, and 48.

Solution:

We divide the numbers by their prime factors until all the numbers become 1.

2	<u>24</u>	<u>36</u>	<u>48</u>
2	<u>12</u>	<u>18</u>	<u>24</u>
2	<u>6</u>	9	<u>12</u>
2	3	9	<u>6</u>
3	<u>3</u>	<u>9</u>	<u>3</u>
3	1	<u>3</u>	1
	1	1	1

Now, multiply all the divisors:

$$\text{LCM} = 2 \times 2 \times 2 \times 2 \times 3 \times 3 = 144$$

Thus, the LCM of 24, 36, and 48 is 144.

HCF

The **Highest Common Factor (HCF)** of two or more numbers is the largest positive integer that divides each of the numbers without leaving a remainder. In other words, it is the greatest number that is a factor of all the given numbers.

Properties of HCF

1. Common Factors:

- A factor of a number is any integer that divides it completely, leaving no remainder. For example, the factors of 12 are 1, 2, 3, 4, 6, and 12.
- For two or more numbers, common factors are those factors that appear in the list of factors of each number. The HCF is the largest of these common factors.

2. Properties of HCF:

- **Commutative Property:** HCF is independent of the order of numbers, so $\text{HCF}(a, b) = \text{HCF}(b, a)$.
 - **Associative Property:** The HCF of three numbers can be found as $\text{HCF}(a, \text{HCF}(b, c))$, and this allows finding the HCF of multiple numbers iteratively.
 - **Divisibility:** The HCF of two numbers divides each number.
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Finding the HCF

There are several methods to find the HCF of two or more numbers:

1. Listing Factors Method

This is the simplest method but is effective only for small numbers. It involves listing all factors of each number and finding the largest common factor.

Example 6: Find the HCF of 18 and 24.

Solution:

Factors of 18: 1, 2, 3, 6, 9, 18

Factors of 24: 1, 2, 3, 4, 6, 8, 12, 24

Common factors of 18 and 24: 1, 2, 3, 6

HCF = 6 (the largest common factor).

2. Prime Factorization Method

In this method, each number is expressed as a product of its prime factors, and the HCF is obtained by taking the lowest power of each common prime factor.

Example 7: Find the HCF of 36 and 60.

Solution:

Prime factorization of 36: $36 = 2^2 \times 3^2$

Prime factorization of 60: $60 = 2^2 \times 3^1 \times 5$

For the HCF, take the lowest power of each common prime:

Power of 2: 2^2

Power of 3: 3^1

HCF = $2^2 \times 3 = 4 \times 3 = 12$

3. Division Method (Euclidean Algorithm)

The Euclidean Algorithm is a fast and efficient method for finding the HCF of two numbers, especially large ones. It involves dividing the larger number by the smaller one and then repeating the process with the remainder until the remainder is zero. The last non-zero remainder is the HCF.

Steps:

1. Divide the larger number by the smaller number and note the remainder.
2. Replace the larger number with the smaller number and the smaller number with the remainder.
3. Repeat the process until the remainder is zero. The last non-zero remainder is the HCF.

Example 8: Find the HCF of 48 and 18.

Solution:

Step 1: $48 \div 18 = 2$ remainder **12**

Step 2: $18 \div 12 = 1$ remainder **6**

Step 3: $12 \div 6 = 2$ remainder **0**

HCF = 6

Finding the HCF of More Than Two Numbers

When there are more than two numbers, the HCF can be found by finding the HCF of two numbers at a time and then using the result to find the HCF with the next number.

Example 9: Find the HCF of 18, 24, and 30.

Solution:

Step 1: Find HCF of 18 and 24 (HCF = 6)

Step 2: Find HCF of 6 and 30 (HCF = 6)

HCF of 18, 24, and 30 = 6

Relationship Between HCF and LCM

The relationship between the Highest Common Factor (HCF) and the Least Common Multiple (LCM) of natural numbers can be used to solve problems efficiently. A key property is that the product of the LCM and HCF of any two numbers is equal to the product of the numbers themselves.

The following formula includes both HCF and LCM:

The product of two numbers = (the two numbers' HCF) x (the two numbers' LCM)

If A and B are the two integers, then:

$$A \times B = \text{H.C.F.}(A, B) \times \text{L.C.M.}(A, B)$$

Example 10: For the numbers 40 and 60 verify the above formula:

Solution:

$$\text{HCF}(40, 60) = 20$$

$$\text{LCM}(40, 60) = 120$$

$$\text{Product of the two numbers} = 40 \times 60 = 2400$$

$$\text{Verifying: } \text{LCM} \times \text{HCF} = 120 \times 20 = 2400$$

Example 11: HCF and LCM of any two whole positive numbers are 35 and 525 respectively. If a whole number is 105, what will be the second whole number?

Solution:

Given:

$$\text{HCF} = 35$$

$$\text{LCM} = 525$$

$$\text{A whole number} = 105$$

Formula used:

$$\text{HCF} \times \text{LCM} = \text{First number} \times \text{Second number}$$

Calculation:

$$35 \times 525 = 105 \times \text{second number}$$

$$\Rightarrow \text{Second number} = (35 \times 525)/105$$

$$\Rightarrow \text{Second number} = 175$$

$\therefore$ The second whole number is 175.

Example 12: If the sum of LCM and HCF of two numbers is 396 and the difference between the LCM and HCF is 324 and the 1st number is 72 then find the second number.

Solution:

Given:

$$\text{LCM} + \text{HCF} = 396, \text{LCM} - \text{HCF} = 324$$

$$\text{First number} = 72$$

Formula used:

$$\text{LCM} \times \text{HCF} = \text{First number} \times \text{second number}$$

Calculation:

$$\text{LCM} + \text{HCF} = 396 \quad \text{----(1)}$$

$$\text{LCM} - \text{HCF} = 324 \quad \text{----(2)}$$

By solving (1) and (2)

$$\Rightarrow \text{LCM} = 360, \text{HCF} = 36$$

$$\text{LCM} \times \text{HCF} = \text{First number} \times \text{second number}$$

$$\Rightarrow 360 \times 36 = 72 \times \text{second number}$$

$$\Rightarrow \text{second number} = 180$$

$\therefore$ The second number is 180.

Example 13: The sum of two 3-digit numbers is 684 and their HCF is 57. If x is the LCM of the numbers, then the sum of the digits of x is:

Solution:

Let two numbers be $57a$ and $57b$.

According to the question

$$57 \times (a + b) = 684$$

$$(a + b) = 684/57 = 12$$

$$(a + b) = 12$$

Hence, the required possible pair of values of a and b which are prime to each other are (1, 11) and (5, 7).

Thus, the required pairs of numbers are

If both numbers are 3-digit numbers then the number will be

$$5a = 5 \times 57 = 285$$

$$7b = 7 \times 57 = 399$$

LCM of 255 and 399

$$\Rightarrow 285 = 5 \times 57$$

$$\Rightarrow 399 = 7 \times 57$$

$$\Rightarrow \text{LCM} = 5 \times 7 \times 57 = 1995$$

$$\Rightarrow \text{Sum of digits of } 1995 = 1 + 9 + 9 + 5 = 24$$

The H.C.F. and L.C.M. of fractions

Special Cases in LCM and HCF

- **Co-prime Numbers:**

When two numbers are co-prime (i.e., their HCF is 1), their LCM is simply the product of the two numbers.

Example:

LCM of 13 and 17 (both prime numbers) is $13 \times 17 = 221$

HCF and LCM of Fractions:

For fractions, the formulas to find the HCF and LCM are:

$$\begin{aligned}\text{HCF of Fractions} &= \frac{\text{HCF of Numerators}}{\text{LCM of Denominators}} \\ \text{LCM of Fractions} &= \frac{\text{LCM of Numerators}}{\text{HCF of Denominators}}\end{aligned}$$

Example 14: Find the HCF and LCM of the fractions $4/9$, $8/15$, and $12/20$

Solution:

HCF of numerators (4, 8, 12) = 4

LCM of denominators (9, 15, 20) = 180

So, the HCF of the fractions is $4/180 = 1/45$

For LCM:

LCM of numerators (4, 8, 12) = 24

HCF of denominators (9, 15, 20) = 1

So, the LCM of the fractions is $24/1$

Example 15: What will be the HCF of $7/9$, $14/185$, and $7/10$?

Solution:

Given

$7/9$, $14/185$, $7/10$

Concept used

HCF of fraction = (HCF of numerator) / (LCM of denominator)

Calculation

HCF of 7, 14, 7 = 7

Prime factors of 9 = 3×3

Prime factors of 185 = 5×37

Prime factors of 10 = 2×5

So, LCM of 9, 185, 10 = 3330

HCF of fraction = $7/3330$

Example 16: Find the LCM of 0.75 and $1/3$.

Solution:

Given:

The given numbers are 0.75 and $1/3$.

Concept Used:

LCM of fraction = LCM of numerators/HCF of denominators

Calculation:

The fraction conversion of 0.75 is $\frac{3}{4}$.

LCM of numerator = LCM of (3, 1) = 3

And, HCF of denominator = HCF of (4, 3) = 1

Now,

LCM of (0.75 and 0.33) = (LCM of (3, 1) / (HCF of (4, 3)) = $\frac{3}{1} = 3$

Example 17: Find the HCF of 1.43, 1.87 and 20.9.

A. 11

B. 1

C. 0.11

D. 110

Solution:

Calculation

HCF of fraction = HCF of numerator / LCM of denominator

$1.43 = \frac{143}{100}$, $1.87 = \frac{187}{100}$ and $20.9 = \frac{209}{10}$

HCF of 143, 187, 209 = 11

LCM of 100, 100, and 10 = 100

HCF of 1.43, 1.87 and 20.9 = $\frac{11}{100}$ or 0.11

Remainders in LCM and HCF Problems

In certain types of problems, we find either the smallest or greatest number that, when divided by a set of divisors, leaves specific remainders. Here's how to approach these problems using LCM and HCF.

1. Finding the Smallest Number that Leaves a Common Remainder

Problem Type: Find the smallest number that, when divided by x , y , and z leaves the same remainder R .

The required number is:

$$\text{Required Number} = \text{LCM of } (x, y, z) + R$$

Example 18: Find the smallest number that, when divided by 5, 7, and 9, leaves a remainder of 3:

Solution:

- **Step 1:** Calculate the LCM of 5, 7, and 9, which is 315.
- **Step 2:** Add the remainder $R = 3$.

Thus, the required number is $315 + 3 = 318$

2. Finding the Greatest Number that Leaves a Common Remainder

Problem Type: Find the greatest number that divides x , y , and z to leave the same remainder R .

To solve this, calculate:

$$\text{HCF of } (x - R), (y - R), \text{ and } (z - R)$$

Example 19: Find the greatest number that divides 20, 50, and 70, leaving a remainder of 4:

Solution:

- **Step 1:** Subtract 4 from each number: $20 - 4 = 16$, $50 - 4 = 46$, $70 - 4 = 66$
- **Step 2:** Find the HCF of 16, 46, and 66, which is 2.

So, the required greatest number is 2.

3. Finding the Greatest Number that Divides Multiple Numbers with Different Remainders

Problem Type: Find the greatest number that divides x , y , and z leaving remainders a , b , and c , respectively.

Calculate:

$$\text{HCF of } (x - a), (y - b), \text{ and } (z - c)$$

Example 20: Find the greatest number that divides 23, 34, and 45, leaving remainders 2, 1, and 3, respectively:

Solution:

- **Step 1:** Subtract each remainder from its corresponding number: $23 - 2 = 21$, $34 - 1 = 33$, and $45 - 3 = 42$.
- **Step 2:** Find the HCF of 21, 33, and 42, which is 3.

Thus, the greatest number is 3.

4. Finding the Smallest Number that Leaves Different Remainders

Problem Type: Find the smallest number that, when divided by x , y , and z leaves remainders a , b and c , respectively.

1. Calculate $(x - a)$, $(y - b)$, and $(z - c)$ and find the common factor K (if these differences are all multiples of K).
2. Required number:

$$\text{Required Number} = \text{LCM of } (x, y, z) - K$$

Example 21: Find the smallest number that, when divided by 4, 7, and 9, leaves remainders 3, 5, and 6:

Solution:

- **Step 1:** Calculate $(4 - 3) = 1$, $(7 - 5) = 2$, $(9 - 6) = 3$. Here, the common factor K is 1.
- **Step 2:** Find the LCM of 4, 7, and 9, which is 252.
- **Step 3:** Subtract K: $252 - 1 = 251$

Therefore, the smallest number is 251.

Example 22: The ratio of two numbers is 3 : 4 and their LCM is 72, then the sum of the two numbers is.

Solution:

Given:

The ratio of numbers = 3 : 4

LCM = 72

Concept Used:

If two numbers are in form of mx , my

Then, $LCM = m \times x \times y$

Calculation:

Let the number be $3x$ and $4x$

LCM = 72

$$\Rightarrow 3 \times 4 \times x = 72$$

$$\Rightarrow 12x = 72$$

$$\Rightarrow x = 6$$

Required sum = $3x + 4x$

$$\Rightarrow \text{Required sum} = 3 \times 6 + 4 \times 6$$

$$\Rightarrow \text{Required sum} = 18 + 24$$

⇒ Required sum = 42

∴ The sum of the two numbers is 42.

Key Takeaways

1. **For smallest numbers with a common remainder:** Add the remainder to the LCM of the divisors.
2. **For the greatest numbers with a common remainder:** Use the HCF of the adjusted values.
3. **For greatest numbers with different remainders:** Find the HCF of each divisor minus its remainder.
4. **For smallest numbers with different remainders:** Subtract the common factor from the LCM of the divisors.

These techniques are effective for solving remainder-based problems in LCM and HCF and are commonly tested in competitive exams.
